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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination – APRIL 2021

MT305S – DIFFERENTIAL EQUATIONS

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Solve : $\sin px \cos y = \cos px \sin y + p$.
2. Write the general form of Cauchy's homogeneous linear equation.
3. Form the partial differential equation by eliminating the arbitrary constants from $z = (x^2+a)(y^2+b)$.
4. Obtain the complete integral of $p^2 + q^2 = npq$.
5. $L(e^{-t}+\sinh 2t) = ?$

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Solve: $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = e^{2x}$.
7. Solve: $4\frac{dx}{dt} + 9\frac{dy}{dt} + 2x + 31y = 0$; $3\frac{dx}{dt} + 7\frac{dy}{dt} + x + 24y = 0$.
8. Solve: $xdx + zdy + (y+2z)dz = 0$.
9. Use Charpit's method to solve: $pxy + pq + qy = yz$.
10. Evaluate : $L^{-1}\left(\frac{s+3}{(s^2+2s+10}\right)$.

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) Solve: $y = 2px + p^2$.

(or)

b) Solve : $(D^2-4D+4)y = x^3e^{2x}$.

12. a) Solve: $x^2\frac{d^2y}{dx^2} + 7x\frac{dy}{dx} + 13y = \log x$.

(or)

b) Solve: $\frac{dx}{dt} + \frac{dy}{dt} - 2y = 2\cos t - 7\sin t$; $\frac{dx}{dt} - \frac{dy}{dt} + 2x = 4\cos t - 3\sin t$.

13. a) Solve: $\frac{dx}{y+zx} = \frac{dy}{-x-yz} = \frac{dz}{x^2-y^2}$.

(or)

b) Solve: $yzp + zxq = xy$.

14. a) Obtain the complete and singular solutions of $z = px + qy + p^2 + pq + q^2$.

(or)

b) Solve the partial differential equation $p^2 - xp - q = 0$ by Charpit's method.

15. a) i. $L^{-1}\left(\frac{1}{s(s+1)(s+2)}\right)$=? ii. Find the inverse Laplace transform of $\log(\frac{1+s}{s})$.

(or)

b) Solve $\frac{d^2y}{dt^2} - 4\frac{dy}{dt} + 5y = 4e^{3t}$ given that $y(0) = 2$ and $y'(0) = 7$.

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